DeepSentinel: Filter-Resistant Latent Sentinels for Ownership Verification in Retrieval-Augmented Image Generation

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Abstract

Retrieval-augmented image generation (RAIG) systems can silently reuse private or licensed image datasets by indexing them as retrieval memories. Existing sentinel-based defenses insert key-bearing images into the protected dataset and later query a suspect system with the secret keys, but visible or lexical keys also give an adaptive dataset thief a simple filtering rule. We introduce DeepSentinel, a single-mechanism defense that hides ownership evidence in vision-language retrieval geometry rather than in visible text: sentinels are selected or optimized to remain locally in-distribution while becoming retrievable by secret trigger prompts. This design turns sentinel removal into a collateral-damage tradeoff, since filters aggressive enough to remove DeepSentinel samples also remove normal high-density data. A synthetic retrieval pilot supports this mechanism; our planned full evaluation tests the claim on CLIP retrieval over MS-COCO and Product-10K and on RAIG pipelines under OCR, outlier, and density-based filtering attacks.

Introduction

Image generation systems increasingly use retrieval to ground outputs in large visual memories. This improves controllability and factual grounding, but it also creates a new dataset governance problem: a model provider can index a private image collection and expose its value through generation without revealing the collection itself. ImageSentinel formalizes this threat for retrieval-augmented image generation and proposes sentinel images that respond to secret random keys [Luo et al.(2025)Luo, Zhao, Cheung, See, and Wan]. The approach is attractive because it enables black-box ownership testing.

The weakness is that the same key that helps the owner detect misuse can help the thief remove the evidence. If sentinel images contain random strings, unusual text, abnormal captions, or outlier embeddings, a preprocessing pipeline can filter them before indexing the stolen dataset. This attack is especially natural for RAIG because dataset builders already run deduplication, quality filtering, OCR filtering, and embedding-based outlier removal.

We propose *DeepSentinel*, a filter-resistant alternative. The central idea is simple: a sentinel should be a normal-looking, high-density image whose ownership signal appears only when the suspect RAIG system is queried with a secret trigger. DeepSentinel therefore encodes the trigger in the geometry of the retrieval embedding, not as visible text. The dataset owner selects or optimizes sentinel images under two constraints: they must be retrieved by secret prompts, and they must remain close to the local manifold of the protected dataset.

Our intended contribution is a focused defense and evaluation framework:

- We formalize adaptive filtering attacks against key-bearing RAIG sentinels.
- We introduce distribution-preserving latent sentinels as a single replacement for visible key images.
- We evaluate whether removing latent sentinels requires high collateral damage to normal images, using both retrieval-only and RAIG settings.

Threat Model

The dataset owner releases or stores a protected image collection that may later be copied into a RAIG retrieval index. The adversary can inspect and preprocess the dataset before indexing. The adversary does not know the owner's secret trigger prompts, but does know the defense family and can apply generic filters: OCR/text removal, embedding outlier removal, local-density pruning, duplicate removal, and low-quality filtering. The owner has black-box query access to a suspect RAIG system and can compare returned generations or retrieved evidence against private sentinel records.

Method

Let $f_I(x)$ be the image encoder used by the retrieval system and $f_T(t)$ be the text encoder for a prompt t. A visible-key sentinel makes x contain or depict a rare key string. DeepSentinel instead constructs a sentinel x_s that satisfies:

$$\mathrm{sim}(f_I(x_s), f_T(t_s)) \geq \tau_{\mathrm{trigger}}, \quad (1)$$

$$d_{\mathrm{local}}(f_I(x_s), \mathcal{D}) \leq \tau_{\mathrm{density}}, \quad (2)$$

where t_s is a secret trigger and d_{local} measures local distributional suspiciousness relative to the protected dataset $\mathcal{D}$.

There are two implementation routes. The lightweight route selects naturally aligned high-density images as latent sentinels. The stronger route starts from high-density images and performs constrained optimization toward the

Density drop	Collateral	Visible	Outlier	DeepSentinel
0.05	0.004	0.000	0.042	1.000
0.10	0.045	0.000	0.000	1.000
0.25	0.201	0.000	0.000	1.000
0.50	0.480	0.000	0.000	1.000

Table 1: Synthetic pilot: sentinel survival under density filtering. Visible-key sentinels are removed by text filtering. Hidden outliers are removed by density filtering. DeepSentinel survives because it is locally in-distribution.

trigger embedding while penalizing visual and embedding deviation from the local neighborhood. The paper will use one of these routes as the final method after the CLIP pilot identifies which is more stable.

Experiments

Pilot. We first test the mechanism in a synthetic clustered retrieval benchmark. Normal images are sampled from dense clusters. Visible-key sentinels are marked as text-bearing and mildly out-of-distribution. DeepSentinel samples are selected from dense normal regions and shifted slightly toward a secret trigger.

Full evaluation plan. The main experiments will use MS-COCO and Product-10K subsets. Baselines include ImageSentinel-style visible key sentinels, conventional invisible watermarking, and no-sentinel controls. Attacks include OCR/text filtering, CLIP outlier filtering, local-density filtering, and deduplication. Metrics are ownership detection true-positive rate at fixed false-positive rate, sentinel survival, normal collateral damage, trigger retrieval hit rate, and non-trigger RAIG utility.

Related Work

RAIG ownership protection is introduced by ImageSentinel [Luo et al.(2025)Luo, Zhao, Cheung, See, and Wan]. Broader related areas include dataset canaries, dataset ownership verification, clean-label backdoors, image watermarking, and adversarial protection for diffusion models. This draft intentionally leaves these related-work citations as **[CITATION NEEDED]** until they are verified programmatically.

Limitations

The current evidence is preliminary. The synthetic pilot validates the filtering tradeoff but does not prove that real CLIP encoders allow artifact-free latent trigger alignment. Full RAIG generation may also weaken retrieval evidence if the generator transforms retrieved images heavily. Finally, a strong adversary who knows the exact encoder and is willing to remove substantial normal data may still reduce detection power.

Conclusion

DeepSentinel reframes sentinel-based dataset protection around a single principle: ownership evidence should be hidden in retrieval geometry rather than visible keys. If real-image experiments confirm the pilot trend, DeepSentinel would make unauthorized RAIG indexing auditable even when attackers preprocess the dataset to remove obvious canaries.

References

[Luo et al.(2025)Luo, Zhao, Cheung, See, and Wan] Luo, Z.; Zhao, Y.; Cheung, K. C.; See, S.; and Wan, R. 2025. ImageSentinel: Protecting Visual Datasets from Unauthorized Retrieval-Augmented Image Generation. arXiv:2510.12119.